The Basic Concepts of Disaster and Disaster Risk

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Disaster Risk and Elements at Risk

How does a risk become a disaster? What are the elements at risk?

Disaster Risk

In understanding disaster risk, it is important to review and understand the following terms: disaster, hazard, vulnerability, capacity, risk and disaster, and their interrelationship.

- ***Disaster***. The. serious disruption of the functioning of society, causing widespread human, material, or environmental losses, which exceed the ability of the affected communities to cope using their resources. Disasters occur when the negative effects of the hazards are not well-managed.
- ***Hazard***. Any phenomenon, substance, or situation, which has the potential to cause disruption or damage to infrastructure and services, people, their property, and their environment.
- ***Capacities***. The resources and skills that people possess, can develop, mobilize, and access, which allows them to have more control over shaping their future and coping with disaster risks.
- ***Vulnerability***. A concept which describes factors or constraints of an economic, social, physical, or geographic nature, which reduces the ability of a community to prepare for and cope with the impact of hazards.
- ***Risk***. The probability that negative consequences may arise when hazards interact with vulnerable areas, people, property, and the environment.
- ***Risk Reduction Measures***. These are various activities, projects, and programs that the communities may identify after assessing and analyzing the risks that they face. These measures are specifically intended to reduce the current risks and prevent future risks in the community.
- ***Disaster Risk*** is the chance or likelihood of suffering harm and loss as a result of a hazardous event. It closely depends upon the exposure of a person or a community to a hazard. For example, people living in low-lying mountainous areas or people living in quarrying, logging, or mining sites. Disaster risk is also defined as the probability of harmful consequences or expected losses (death, injuries, property, livelihoods, economic activity disrupted or environment damaged) resulting from interactions between natural or human-induced hazards and vulnerable conditions. This can be expressed as:

$$\boldsymbol{Risk = Chance \times Loss.}$$

The output of risk analysis is usually an estimation of the risk scenarios. The potential disaster losses, in lives, health status, livelihoods, assets, and services, which could occur to a particular community or a society over some specified future time. The definition of disaster risk reflects the concept of disasters as the outcome of continuously present conditions of risk. Therefore, disaster risk comprises different types of potential losses which are often difficult to quantity. Nevertheless, with the knowledge of the prevailing hazards and the patterns of population and socioeconomic development, disaster risks can be assessed and mapped, in broad terms at least (access it using this link: http://www.unisdr.or/we/inform/terminology).

Risk may also depend on exposure to the consequences of uncertainty or potential deviations from what is planned or expected disruption to everyday life following the formula:

$$\boldsymbol{Risk = Probability \times Loss.}$$

Disaster risk can be expressed as a function of hazard × vulnerability or a function (hazard, exposure, vulnerability). How one copes depends on capacity and readiness to respond to an emergency and crisis. The equation,

$$\boldsymbol{Disaster\ Risk} = \boldsymbol{Hazard} \times \boldsymbol{Elements\ of\ Exposure} \times \boldsymbol{Vulnerability},$$

emphasizes particularly the physical aspects of vulnerability.

Risk is the anticipated or potential consequences of a specific hazard interacting with a specific community at a specific time. Risks are a combination of probability (the likelihood of a hazard occurring) and of the consequences or outcomes for the community if exposed to the hazard.

Risk results from the interaction of the three functions namely hazard, vulnerability, and exposure. Beyond expressing a possibility of physical harm, it is crucial to recognize that risks are inherent or can be created or exist within social systems. It is important to consider the social contexts in which risks occur and that people do not necessarily share the same perception of risk and their underlying causes. Risk is also a function of probability while we would all be highly vulnerable when a large meteor strikes the Earth, the probability of its occurrence is so low that the relative risk is also low. This demonstrates that risk is a combination of probability (the likelihood of a hazard occurring) and of the consequences or outcomes for the community and families if exposed to the hazard. At the local and community level, disaster risk is easily understood as the likelihood of a particular hazard occurrence and its probable damaging consequences for people and property. The bigger the vulnerability, the bigger the disaster risk and the bigger the Capacity, the smaller the disaster or,

$$\boldsymbol{Disaster\ Risk} = \boldsymbol{Hazard} \times \boldsymbol{Vulnerability} - \boldsymbol{Capacity} = \frac{\boldsymbol{Hazard} \times \boldsymbol{Vulnerability}}{\boldsymbol{Capacity}}.$$

On the other hand, communities shall take into account the disaster risk assessment to lessen the adverse impacts of disaster risk. According to UNISDR, disaster risk assessment is a participatory process to assess the hazards, vulnerabilities, and capacities of a community. Through hazard assessment, the likelihood of the occurrence, the severity, and duration of various hazards are determined.

The vulnerability assessment identifies what elements are at risk and the causes of their vulnerable conditions. The households and groups that are most exposed to a hazard are identified. The assessment takes into account the physical, geographical, economic, social, and political factors that make some people vulnerable to the dangers of a given hazard.

In the capacity assessment, the community's resources and coping strategies are identified. The result of the disaster risk assessment is a ranking of the disaster risks of the community as the basis of planning for risk reduction. After the completion of hazard, vulnerability, and capacity assessment, it would be essential to conduct a risk analysis. The risk analysis will enable the community and the local authorities to understand the potential impact of various hazard events. Risk analysis involves the development of risk scenarios based upon the information about hazards, their frequency and intensity, and the elements-at-risk. A hazard can impact many elements-at-risk in different manners. During risk analysis we identify what kind of impact a hazard will have on various at-risk elements; e.g. people, houses, buildings, roads, schools, etc. It also identifies the extent of the impact; e.g. death toll, number of injuries, and extent of the damage.

Certain communities may be exposed to more than one hazard. In such areas, it will be important to identify the potential losses from the various kinds of hazards. Different hazards may have a differential impact on various elements-at-risk. For example, earthquakes can be more dangerous in terms of killing and harming people than floods. On the other hand, earthquakes have very little impact upon crops, while floods have a very severe impact, depending upon the cultivation season.

As a result of the analysis of risks, risk statements, or risk scenarios can be prepared to indicate the impact upon various at-risk elements from multiple hazards. Visual risk maps can also be

produced based on hazard maps with information on the level of physical vulnerability. The risk map showing the results of both hazard and vulnerability analysis is regarded as the most important tool in risk analysis. In the risk maps, hazard maps are superimposed on maps of physical aspects of vulnerability.

Knowing the risks and taking actions by identifying, assessing, and monitoring risks will save lives.

Elements-at-Risk

What are the elements-at-risk? Is it the right of at-risk people to know about the risks they face?

The people, properties, systems, or other elements present in hazard zones are subject to potential losses that can be considered as exposed to risks or elements-at-risk. Measures of exposure can include the number of people or types of assets in an area. These can be combined with the specific vulnerability of the exposed elements to any particular hazard to estimate the quantitative risks associated with that hazard in the area of interest.

Community as an Element-at-Risk

A community is one of the elements-at-risk. A community can be defined geographically (a cluster of households, a small village, or a neighborhood in a town), people with shared experience (a particular interest groups, ethnic groups, professional groups, language groups, particular hazard-exposed groups), by sector (farmers, fisherfolk, business sector) and can be referred to groups that are affected by and can assist in the mitigation of hazards and reduction of vulnerabilities. If a community is not prepared before, during, and after a disaster, members of the community will become at risk. If the community is familiar with ways of coping and precautionary measures, then the disruption by a disaster can be reduced (Sampath, 2001).

In the capacity assessment, the community's resources and coping strategies are identified. The result of the disaster risk assessment is a ranking of the disaster risks of the community as a basis of planning for risk reduction. Disaster preparedness covers activities to enhance the ability to predict, respond to, and cope with the effect of a disaster. It includes precautionary activities by households, communities, and organizations to react appropriately during and following the event.

Essential Components in Determining Risk

The following are essential components in the determination of risk, each of which should be separately quantified:

- ***Hazard occurrence probability*** is the likelihood of experiencing a natural or technological hazard at a given location or region. Quantifying hazard probability involves assessing not only the probability of occurrence but the probability of magnitude.
- ***Elements-at-risk***. Identifying and making an inventory of people or school buildings or other elements that would be affected by the hazard if it occurs, and when required, estimating their economic value.
- ***Vulnerability of the elements-at-risk***. Effect on school buildings or schoolchildren or other elements if they experienced some levels of hazard impact. Vulnerability is the relationship between the severity of hazard impact and the degree of damage caused. Each element is affected differently by the hazards of different severity.

Loss Management

These are the pre- and postdisaster actions designed to keep the losses at the minimum in human, structural, and economic aspects.

- ***Predisaster loss management*** area activities focusing on reducing the community vulnerability to hazards. Actions include improving the resistance of physical structures such as school buildings, developing improved safety plans for the occupants, and increasing/diversifying the network of social support mechanisms available to communities in threatened areas.
- ***Postdisaster loss management*** focuses on improving the emergency response and broadening the range of support given to victims that include the facilitation of relief delivery and stimulating a rapid recovery.

Effects of Disasters

The effects of disasters can be categorized:

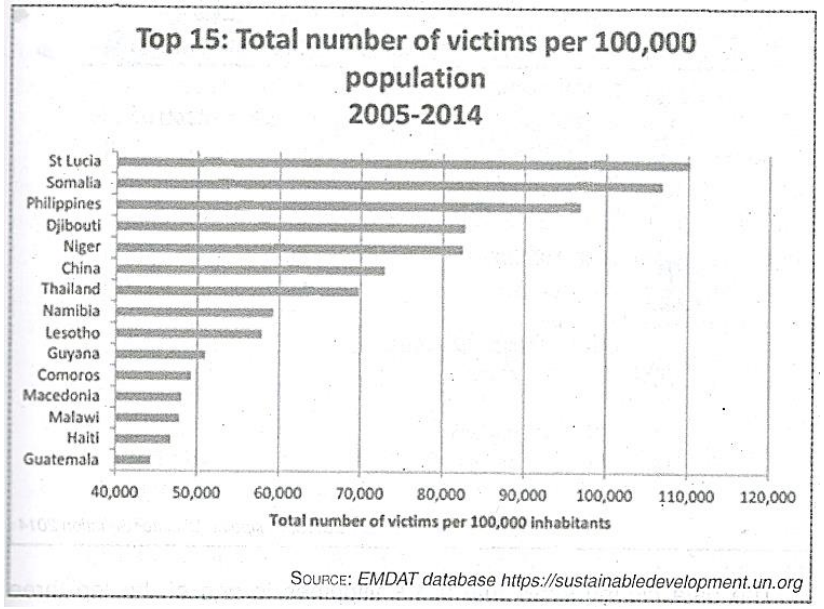
1. ***Human/Biological***

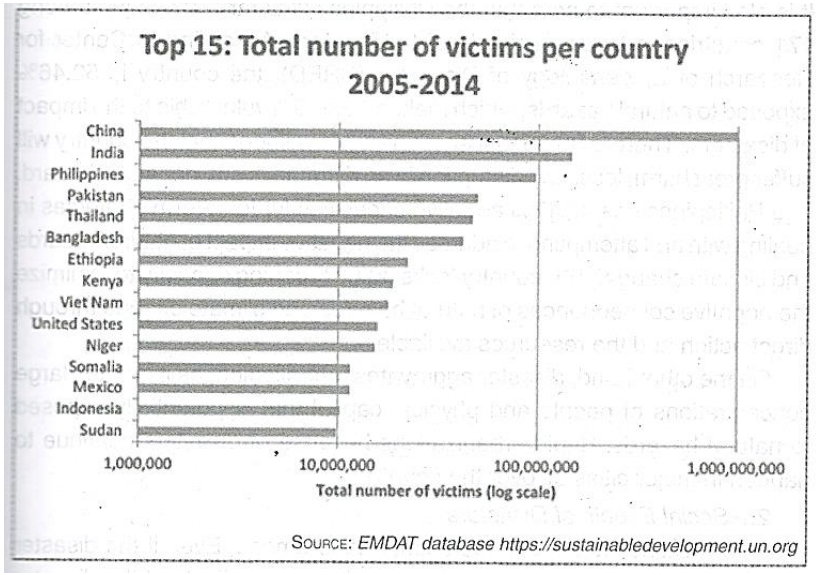
During and after disasters, a high incidence of disease within the community happened. Infectious diseases are the most usual form of an epidemic disease because of the congested evacuation areas. It is easier to "infect" and therefore to suddenly affect the population/evacuees which means they produce unexpected illness or death.

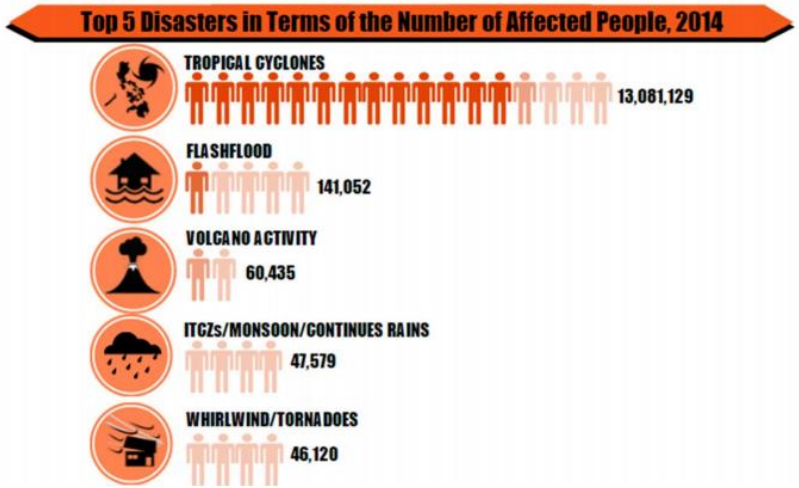
Human capital accumulation positively affects economic growth in most circumstances. If the technology embodied in the physical capital of an economy is sophisticated, then investment in human capital yields more educated, competent workers, and workers who can learn and apply new concepts quickly are more productive than workers who cannot. Investment in human capital, however, is not a sufficient condition for economic growth. A country with insufficient opportunities for workers may not have the physical capital or types of jobs necessary to justify the investment in human capital.

Disasters affect human capital accumulation in several ways. Initially, disasters substantially reduce human capital only if there is a substantial loss of life. Between 1970 and 2001 (excluding droughts), only three natural disasters resulted in the deaths of over 100,000 people and only nineteen (19) natural disasters that resulted in the deaths of over 10,000 people (Arulsamy, 2011).

In the last decades, reports on the effects of disasters have been increasing. There is a significant effect on long term indicators as manifested in the Human Development Index. The data clearly show that the Philippines is one of the top three countries that are vulnerable to disasters in terms of loss of people's lives. It is also important to note that the Philippines was ranked second among 171 countries in terms of risk level to disasters. According to the Center for Research of Epidemiology of Disasters (CRED), the country is 52.46%







Source: http://www.cdrc-phil.com/wp-content/uploads/2015/08/2014-Philippine-Disaster-Report.pdf

exposed to natural hazards, which makes it 53.85% vulnerable to the impact of disasters. There is 33.35% susceptibility of a likelihood that the country will suffer great harm, loss, and disruption in an extreme event, or natural hazard. The Philippines has 48.17% adaptive capacities for long-term strategies in dealing with and attempting to address the negative impact of natural hazards and climate change. The country lacks 80.03% coping capacity to minimize the negative consequences of natural hazards and climate change through direct action and the resources available.

On the other hand, disaster aggravates urbanization, as it creates large concentrations of people and physical capital that is potentially exposed to natural hazards. Rapid urbanizations because of disasters continue to happen in major cities all over the country.

2. ***Social Effects of Disasters***

Disasters affect people first and foremost. Even if the disaster does not kill anyone, emotional and physical effects of the disaster on the population still harm people well after the actual disaster, Some survivors will likely suffer permanent physical disabilities and psychological conditions, such as posttraumatic stress disorder (PTSD). Survivors who suffer from psychological conditions and disabilities will not be as productive as they were before the disaster, which could affect growth if the disasters affect enough people. The effects of the disasters on the mental state of the people are important. A depressed, demoralized, and traumatized population is less productive' than a population who successfully endured a disaster. Like the case of typhoon Yolanda survivors. According to the reports of the local government units in the province of Leyte, the majority of their youth and children were engaged in alcohol, and drug abuse, as well as teenage pregnancy, were rampant after the disaster. This is a manifestation of the postdisaster situation in October 2015 data revealed by the Leyte Provincial Disaster Risk Reduction and Management Officer-In-Charge. Children and young people are emotionally disturbed, thus they experience recurrent nightmares, regression in developmental achievements, increased dependency on parents and guardians, decreased appetite, clinging, exaggerated startled response, irritability, and school problems.

The after-effects of a disaster can disrupt the lives of the people.

3. ***Physical/Material***

Before the disaster, poor people suffer more from crises than people who are richer because they have little or no savings, less income or production options, and limited resources. After disasters, the more these people become poorer and vulnerable. Damage to physical and material has a significant impact on our society. Examples are collapsed buildings, damaged houses, poor evacuation sites, and the like.

4. ***Psychological***

The psychological responses of the survivors can range from transient mild stress reactions to the more severe and prolonged consequences of PTSD. These responses are influenced by the gender, developmental stage, inherent resilience, social support, and the level of exposure of the person to the trauma. Exposure to traumatic and violent

events results in expressions of generalized fear, anxiety, and depression.

Reports revealed that during the first few weeks and months after the disaster, survivors are lost and disoriented. Victims suffer from anxiety, despair, grief, severe sleep disturbances, and nightmares. Anxieties are manifested in phobias and obsessions about wind, rain, loud sounds, generalized fears, and other reminders that the disaster could recur. The victim's emotional outbursts to feelings of numbness affected their routine activities like work and recreation. Extreme effects are major depression, nervous breakdown, and a lot of survivors were diagnosed having a psychological disorder.

5. ***Economic***

The impact of a natural disaster may also reflect a sense of inequalities. During the aftermath of a disaster, the poor who suffer from income fluctuations and have limited access to financial services may be more prone to reduce consumption. Also, there are some nonpoor or and not prepared or insured against those risks, may fall into the poor category as a consequence of decapitalization when coping with the hazards or disaster impacts.

Moreover, vulnerability to natural disasters is a complex issue, as it is determined by the economic structure, the stage of development, prevailing social and economic conditions, coping mechanism, risk assessment, frequency and intensity of disasters, etc. The impact on the poor could be losing access to some basic services, reversals in accumulation of physical and human capital, and perhaps an increase in child labor, prostitution; and criminal activities.

Disaster, extreme poverty, and increasing inequality have important consequences in achieving our Sustainable Development Goals or SDGs. Inequality can slow down and can even hamper development toward the reduction of poverty and risk reduction.

- ***Direct Costs***: Physical damage; including that of productive capital and stocks (industrial plants; standing crops, inventories), economic infrastructure (roads, electricity supply, etc.) and social infrastructure (homes, schools; hospitals)
- ***Indirect Costs***: Disruptions to the flow of goods and services lower output from damaged or destroyed assets and infrastructure and the loss of earnings as income-generating opportunities are affected. Disruption of the basic services, such as telecommunication or water supply can have far-reaching implications. The impacts are severe in low-income countries. For example, in 1995 typhoons in the Philippines caused damage of US$ 350 million.

6. ***Environmental Degradation***. The effects of climate change manifest not only in the environment's degradation. Global warming also affects energy, agriculture, health, water, and marine resources. The downstream effects of climate change would make an already bad situation worse. Extreme changes in the weather and the rise of carbon dioxide in the atmosphere would reduce the harvests of farmers, cause habitat degradation and species loss, epidemics, and diseases, kill corals, and disrupt carbonate chemistry making shell and bone formation difficult. Climate change also aggravates environmental hazards. In the Philippines, disasters, whether climate-induced or not, add up to the already impoverished situation of the majority of Filipino families who are living below the poverty line. The harmful effects of climate change and the disasters bear heavily on the most vulnerable or marginalized segments of the Philippine population, especially the poor farmers and indigenous communities. Large-scale foreign mining on the other hand results in the destruction of the environment, loss of livelihood, displacement of communities, and cause health complications.

7. ***Political Impacts of Disasters***. Disaster represents extreme shocks to the political system of the country, increasing the number of citizen demands while simultaneously reducing the government's response capabilities. The political impact of disaster is less easy to pin down than the direct economic impacts of disaster. Political impacts of disaster are often determined by pre-disaster political context. Communities or countries that are devastated by disasters often experience significant political transformations soon afterward (e.g. appointment of new DRR officers, new policies, and systems at the national and local level, etc.). Such changes are more likely to occur in places with an inequitable distribution of wealth and a history of sociopolitical conflict. Disasters encourage the organization of victims/survivors, highlighting the commonalities between them, and encourage them to act in interest of the public good, demand for their rights as right-bearer and questioned the actions of the duty-bearers. Such mobilization eventually spills into the political sphere, increases people's voice in the political system and often leads to a change in the political system or increase the political awareness of the general public. In other words, disasters strengthen civil society, challenge or change the existing governance patterns, and thus contribute to a process of democratization.

The Philippines is an archipelago with two pronounced seasons which are categorized as wet and dry. The month of June starts the wet season and reaches until February with the rest of the months being the dry season. Typhoons, strong rains, and floods abound during the wet season along with incidents of landslides. Scorching heat is experienced during the dry season as temperatures rise with the passing of each month. In some cases, agricultural production is affected as water sources dry up.

The archipelagic characteristic of the country makes coastal areas susceptible to floods and saltwater intrusion due to rising tides from the seas and ocean. Being in the Western Pacific Basin, the Philippines is a typhoon path with at least 20 typhoons experienced yearly. The content of the report of the Center for Research of Epidemiology of Disasters (CRED) every year impacted the whole world as it is presented a disturbing and alarming state of adverse impacts of disasters and hazards and grim prospects for the future given its effects. Adverse effects to agriculture and food production, destruction of environment and biodiversity, extinction of plant and animal species, war, water shortage, worsened poverty situation, flooding, frequency of extreme weather-related events, and epidemics are just some examples of effects of disasters.

The increasing concentration of greenhouse gases in the atmosphere has caused the development of a large hole in the ozone layer, the Earth's protective layer against the intense heat of the sun. Global temperatures have increased significantly over the past few decades causing alterations in weather patterns and melting of icecaps that subsequently increased sea levels worldwide.

During the period 1971—2000, the temperature in the Philippines has been observed to have increased by $\mathbf{0.14}$℃. It has also been observed that mean rainfall and the number of rainy days has increased since the 1980s. The effects and impacts of disasters worldwide and that of the Philippines have already been felt through various disaster events in the country. It will continue to be felt in the decades to come. Above all, disaster is seen to impact greatly the already vulnerable and poor section of the society. As it is, their coping capacities are already very low and further aggravated by the poverty situation and worsening social crises of social and gender inequality, insufficiency of social services, and corruption in government.

Disasters are emergencies that cannot be handled by those affected without outside assistance. They are caused by natural or man-made events wherein communities experience severe danger and incur loss of lives and properties causing disruption to their social structure and to all or some of the affected communities' essential functions. Disasters are inevitable. They are caused by unsustainable development that has not taken account the possible hazard impacts in a particular location. They can be less damaging if the community has a better understanding of locally-experienced hazards and implements preventive or mitigating measures against them.

Reference:

Lanada, M.I.B., Melegrito, M.L.F., & Mendoza, D.J. (2016). *Building resilient communities: Disaster readiness and risk reduction.* Phoenix Publishing House